SphereX

Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer

The Spectro-Photometer for the History of the Universe, Epoch of Reionization and Ices Explorer (SPHEREx) mission will provide an all-sky spectral survey. Over a two-year planned mission, the SPHEREx Observatory will collect data on more than 450 million galaxies along with more than 100 million stars in the Milky Way in order to explore the origins of the universe.

Launched on - March 11, 2025

Objective - Survey the sky in optical as well as near-infrared light

https://irsa.ipac.caltech.edu/Missions/spherex.html?utm_source=chatgpt.com

Why does Time Stamps Matter ?

An observation timestamp tells us when a particular measurement was taken. We need it to determine whether two observations happened at different times and whether a source could have changed between them.

IR Ranges :-

1) 0.7 - 1.1 um -> R = 39

Near the shortest wavelengths SPHEREx observes. Useful for studying stellar and galaxy light, including visible-light emission shifted into the near-infrared.

2) 1.1–1.6 μm -> R = 41

Extends the near-infrared spectrum, helping characterize stars and galaxies and their spectral energy distributions

3) 1.6–2.4 μm -> R = 41

Provides additional spectral information about stars, galaxies, and distant sources. It contributes to estimating galaxy distances and studying the distribution

Of cosmic light.

4) 2.4–3.8 μm -> R = 35

Particularly relevant to molecular and interstellar-ice research, including the broad

water-ice absorption feature near 3.0 μm.

5) 3.8–4.4 μm -> R = 112

Offers finer spectral detail in the longer-wavelength infrared. Useful for studying features such as carbon dioxide ice absorption near 4.27 μm.

6) 4.4–5.0 μm -> R = 128

Has the highest nominal resolving power in your table. It helps investigate longer-wavelength molecular-ice features, including carbon monoxide ice near 4.67 μm.

FRONT-END : -

This is one of the most useful things you can give the frontend developer.

Field	Why it's useful
Object or candidate ID	Identifies the source being investigated.
Sky coordinates	Shows where the source is located.
Observation dates	Establishes when measurements were collected.
Wavelength or spectral channel	Identifies which measurements are being compared.
Measured displacement or brightness change	Describes the observed difference.
Measurement uncertainty	Indicates how precisely the difference is known.
Data-quality indicators	Highlights potential measurement problems.
Scientific interpretation	Explains what the evidence supports.

Validation status	Distinguishes plausible candidates from inconclusive or likely-artifact cases.

BACKEND : -

The backend should retrieve or receive the following information for each observation.

Data field	Why it is required
Observation image	Contains the astronomical sources to analyze.
Observation timestamp	Establishes when the measurement was taken.
Sky-coordinate metadata (WCS)	Converts image pixels into celestial coordinates.
Wavelength information	Helps ensure that observations being compared are compatible.
Source coordinates	Identify the measured position of each source.
Positional uncertainty	Helps determine whether a measured shift is meaningful.
Data-quality flags	Identify potentially unreliable measurements.
Calibration and observation metadata	Help assess whether measurements can be compared fairly.